MATHEMATICS

REVISION PLAN-3

Target : JEE (Advanced)-2023

DPP
DAILY PRACTICE PROBLEM
DPP NO. # 07 (Advanced)

Max. Time : 129 Minutes **Total Marks-166**

Q. 01 to 06	**Comprehension**	**('–1' negative marking)**	**[3 Marks,3 Min.][18,18]**
Q. 07 to 26	**Multiple choice objective**	**('–1' negative marking)**	**[4 Marks,3 Min.][80,60]**
Q. 27 to 28	**Match the column**	**(no negative marking)**	**[4 Marks,3 min.][8,6]**
Q. 29 to 43	**Numerical Type**	**('–1' negative marking)**	**[4 Marks,3 Min.][60,45]**

DPP Syllabus : Quadratic Equation, Matrices and Determinants

Comprehension # 1 (Q.No. 1 to 2)

If α, β are the roots of equation $(K + 1)x^2 - (20K + 14)x + 91K + 40 = 0$; $(\alpha < \beta)$, $K > 0$, then answer the following questions

1. The nature of the roots of this equation is :
(A) imaginary (B) real and distinct (C) equal real roots (D) None of these

2. The smaller root (α) lie in the interval
(A) (4, 7) (B) (7, 10) (C) (10, 13) (D) None of these

Comprehension # 2 (Q.No. 3 to 4)

Let $\Delta = \begin{vmatrix} -bc & b^2+bc & c^2+bc \\ a^2+ac & -ac & c^2+ac \\ a^2+ab & b^2+ab & -ab \end{vmatrix}$ and the equation $px^3 + qx^2 + rx + s = 0$ has roots a, b, c where $a, b, c \in R^+$

3. The value of Δ is equal to
(A) $\frac{r^2}{p^2}$ (B) $\frac{p^2}{r^2}$ (C) $\frac{r^3}{p^3}$ (D) $\frac{p^3}{r^3}$

4. If $\Delta = 27$ and $a^2 + b^2 + c^2 = 3$, then
(A) $3p + 2q = 0$ (B) $4p + 3q = 0$ (C) $3p + q = 0$ (D) none of these

Comprehension # 3 (Q.No. 5 to 6)

Consider two 3×3 matrices A and B satisfying $A = adj(B) - B^T$ and $B = adj(A) - A^T$. Also given that A is a non singular matrix. (where P^T denotes transpose of matrix P).

5. $|A| + |B|$ is equal to
(A) 4 (B) 8 (C) 16 (D) 128

6. $AB + BA$ is equal to
(A) 16I (B) 2I (C) 4I (D) 8I

7. Set of equation $x^2 - b_n x + \underbrace{111.....1}_{n-\text{times}} = n$, n = 2, 3, 4, ... 9 and $b_n \in N$, have integer roots, Which of the following are INCORRECT ?

(A) Common root of these equations can be 9

(B) No three of these equations have a common root.

(C) A root of an equation with n = 4 can be 123.

(D) A root of an equation with n = 4 is 1234.

8. If $x^4 - (a + 1)x^3 + x^2 + (a + 1)x - 2 = 0$, $a \in R$ have atleast two positive real roots, then 'a' can be in the interval

(A) $a \in \left(-\infty, -1-2\sqrt{2}\right]$ (B) $a \in \left[2\sqrt{2}-1, \infty\right)$

(C) $a \in \left[-15, -1-2\sqrt{2}\right]$ (D) $a \in \left[2\sqrt{2}-1, 15\right]$

9. Let the roots of the equation $x^3 + ax + a = 0$. $a \in R - \{0\}$ be α, β and γ such that $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\gamma} + \frac{\gamma^2}{\alpha} + 8 = 0$ and $\alpha < \beta < \gamma$. then

(A) $\alpha = -1-\sqrt{5}$ (B) $\beta = 1 - \sqrt{5}$ (C) $\gamma = 1 + \sqrt{5}$ (D) $a = -8$

10. If $\begin{vmatrix} x^2-5x+3 & 2x-5 & 3 \\ 3x^2+x+4 & 6x+1 & 9 \\ 7x^2-6x+9 & 14x-6 & 21 \end{vmatrix} = ax^3 + bx^2 + cx + d$, then which of the following is/are correct ?

(A) a = 0 (B) b = 0 (C) c = 0 (D) d = 0

11. A is a square matrix of order 3. A, A^{-1}, A^T all have the same value of determinant. Also (adj A) = A^T, then

(A) det $(A^{-1}) = 1$

(B) $(ABA^T)^2 = A^2B^2(A^T)^2$; B is a 3 × 3 matrix

(C) $\det((ABA^T)^{1000}) = \det(B) \Rightarrow \det(B) = 0, 1$ or -1

(D) $(ABA^T)^{-1} = AB^{-1}A^T$, if B is an invertible 3 × 3 matrix

12. Let $P = \begin{bmatrix} 5a^2+2bc & 6 & 8 \\ 13 & 8b^2-10ac & -9 \\ 7 & 5 & 25c^2 \end{bmatrix}$, $Q = \begin{bmatrix} a^2+6ab & 3 & 5 \\ 12 & -b^2 & 6 \\ 1 & 4 & 17bc \end{bmatrix}$, a, b, c $\in$ N. if trace(P) = trace(Q) and a, b, c are sides of ΔABC with BC = a, CA = b and AB = c, then $-120 \cos A$ is

(trace (P) = sum of principal diagonal elements of matrix P)

(A) Prime Number (B) Even Number (C) Odd Number (D) Positive Number

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13. If A is a square matrix of order n such that $|\text{adj (adj A)}| = |A|^9$, then

(A) n is composite
(B) n! has 8 natural divisors
(C) $|\text{cof-A}| = |A|^3$
(D) A can have maximum 9 distinct elements

14. Let A be a square matrix of order 3 which satisfies the equation $A^4 - 8A^3 + 24A^2 - 37A + 16I = 0$ and $B = A - 2I$ be another matrix. If Det A = 5 and Det B > 0, A^{-1} represent inverse of matrix A, then

(A) $\text{Det}(\text{adj}(I - 2A^{-1}))$ is equal to 1
(B) $\text{Det}\left(\text{adj}\left(\frac{B}{5}\right)^{-1}\right)$ is equal to 625
(C) $\text{Det}(I + 2B^{-1})$ is equal to 1
(D) Det B = 25

15. The roots α, β of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ lie in the interval [0, 1], then?

(A) the maximum value of $\frac{(a-b)(2a-b)}{a(a-b+c)}$ is 3

(B) the minimum value of $\frac{(a-b)(2a-b)}{a(a-b+c)}$ is 2

(C) if $\frac{(a-b)(2a-b)}{a(a-b+c)}$ is at its maximum value, then $\alpha^2 + \beta^2$ can be 2

(D) if $\frac{(a-b)(2a-b)}{a(a-b+c)}$ is at its maximum value, then $\alpha^2 + \beta^2$ can be 1

16. If $A = \begin{pmatrix} \alpha & 0 \\ 2 & \alpha \end{pmatrix}$ and $\det(2A^2 - 2A) = 144$, then possible correct statement is (for real part)

(A) tr (A) = 6
(B) det (A) = 9
(C) det (A) = 4
(D) tr (A) = 1

17. For $\theta \in [0, 2\pi]$, let $\Delta(\theta) = \begin{vmatrix} \cos\theta & \cos 2\theta & \cos 3\theta \\ \cos 2\theta & \cos 3\theta & \cos 4\theta \\ \cos 3\theta & \cos 4\theta & \cos 5\theta \end{vmatrix}$, then

(A) $\Delta\left(\frac{5\pi}{12}\right) = 0$
(B) $\Delta\left(\frac{7\pi}{12}\right) = -1$
(C) $\Delta\left(\frac{2\pi}{3}\right) = 0$
(D) $\Delta\left(\frac{3\pi}{4}\right) = \sqrt{3}$

18. Let $\omega \neq 1$ be cube root of unity and let $P = p_{ij}$ be n × n matrix such that $p_{ij} = \omega^{i+j}$. Then $P^2 \neq O$ when n is

(A) 57
(B) 55
(C) 58
(D) 56

19. Let $\alpha = \pi/5$ and $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, then $B = A + A^2 + A^3 + A^4$ is

(A) singular
(B) non-singular
(C) skew-symmetric
(D) |B| = 1

20. Let A, B be two 2 × 2 real commutative matrices, such that det $(A^2 + AB + B^2) = 0$, then (ω is non real cube root of unity and $A^3 \neq B^3$)

(A) det $(A - \omega B) = 0$
(B) det $(A - \omega^2 B) = 0$
(C) det $(A + B) = 0$
(D) det $(A - B) = 0$

21. If exactly two integers lie between the roots of equation $x^2 + ax - 1 = 0$. Then integral value(s) of 'a' is/are:

(A) –1
(B) –2
(C) 1
(D) 2

22. For the quadratic polynomial $f(x) = 4x^2 - 8ax + a$, then statements(s) which hold good is/are

(A) There is only one integral 'a' for which f(x) is non-negative $\forall\ x \in R$
(B) For $a < 0$, the number zero lies between the zeroes of the polynomial
(C) $f(x) = 0$ has two distinct solutions in (0, 1) for $a \in \left(\frac{1}{4}, \frac{4}{7}\right)$
(D) None of these

23. If a, b are two numbers such that $a^2 + b^2 = 7$ and $a^3 + b^3 = 10$, then :

(A) The greatest value of $|a + b| = 5$
(B) The greatest value of $(a + b)$ is 4
(C) The least value of $(a + b)$ is 1
(D) The least value of $|a + b|$ is 1

24. Let $|a| < |b|$ and a, b are the root of the equation $x^2 - |\alpha|\, x - |\beta| = 0$. If $|\alpha| < b - 1$, then the equation $\log_{|a|}\left(\frac{x}{b}\right)^2 - 1 = 0$ has at least one

(A) root lying in $(-\infty, a)$
(B) root lying in (b, ∞)
(C) negative root
(D) positive root

25. Let A and B be two matrices different from I such that AB = BA and $A^n - B^n$ is invertible for some positive integer n. If $A^n - B^n = A^{n+1} - B^{n+1} = A^{n+2} - B^{n+2}$, then

(A) I – A is singular
(B) I – B is singular
(C) A + B = AB + I
(D) (I – A) (I – B) is non-singular

26. If there are three square matrix A, B,C of same order satisfying the equation $A^2 = A^{-1}$ and let $B = A^{2^n} A^{2n}$ & $C = A^{2^{(n-2)}}$, then which of the following statements are true?

(A) det. $(B - C) = 0$
(B) $(B + C)(B - C) = 0$
(C) B must be equal to C
(D) none of these

27. **Column -1** **Column – II**

(A)	If $x^2 + x - a = 0$ has integral roots where $a \in [0, 20]$ then sum of possible values of 'a' is	(p)	2
(B)	If the equation $ax^2 + 2bx + 4c = 16$ has no real roots and $a + c > b + 4$, then sum of integral values of c is ($c \in (6, 10)$)	(q)	40
(C)	If the equation $x^2 + 2bx + 9b - 14 = 0$ has only negative roots, then number of prime values of b is (where $b < 10$)	(r)	24
(D)	The largest positive terms of the HP whose first two terms are $\frac{2}{5}$ and $\frac{12}{23}$ is k, then k =	(s)	6

28. Let A and B be two non-singular matrices such that $(AB)^k = A^kB^k$ for three consecutive positive integral values of k.

Column-I		**Column-II**	
(A)	ABA^{-1}	(p)	A^2
(B)	BAB^{-1}	(q)	B
(C)	AB^2A^{-1}	(r)	A
(D)	BA^2B^{-1}	(s)	B^2

29. Let f be a function such that $f : R \to R$, $f(x) = \dfrac{3x^2 + mx + n}{x^2 + 1}$ and the value of f(x) is $[-4, 3)$, then the value of $m^4 + n^4$ is equal to

30. Let α,β,γ be distinct real numbers such that

$a\alpha^2 + b\alpha + c = (\sin\theta)\alpha^2 + (\cos\theta)\alpha$

$a\beta^2 + b\beta + c = (\sin\theta)\beta^2 + (\cos\theta)\beta$

$a\gamma^2 + b\gamma + c = (\sin\theta)\gamma^2 + (\cos\theta)\gamma$ (where $a, b, c, \in R$.)

Then the maximum value of the expression $\dfrac{a^2 + b^2}{a^2 + 3ab + 5b^2}$ is

31. Let a, b, c be real numbers such that $a < 3$ and all zeros of the polynomial $x^3 + ax^2 + bx + c$ are negative real number. Find sum of possible positive integral values of $b + c$.

32. Let α and β are roots of $x^2 - 17x - 6 = 0$ with $\alpha > \beta$. If $a_n = \alpha^{n+2} + \beta^{n-2}$ for $n \geq 5$, then the value of $\frac{a_{10} - 6a_8 - a_9}{a_9}$

33. Let p be an integer such that both roots of the equation $5x^2 - 5px + (66p - 1) = 0$ are positve integers. Then the value of $\left[\frac{p}{10}\right]$ is equal to ([.] denotes greatest integer function)

34. Number of ordered pair(s) (x, y) that satisfies the inequation $y - |x| \geq 1 + \sqrt{x^2 + y^2 - 1}$ is/are

35. If $a^2 + b^2 + c^2 + ab + bc + ca \leq 0$ then $\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$ is

36. Let A and B are two non singular matrices such that $B \neq I$, $A^5 = I$ and $AB^2 = BA$, then the least value of n for which $B^n = I$ is

37. If AB = BA = I such that $A = \left[a_{ij}\right]_{2\times2}$, $B = \left[b_{ij}\right]_{2\times2}$, $a_{ii} = b_{ii}$, $a_{12} = -b_{12}$, $a_{21} = -b_{21}$ and a_{ij} are coprime natural numbers, then $|a_{21} - a_{12}| = \ldots..$

38. Let $A = \begin{bmatrix} 1 & \frac{3}{2} \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ and $C_r = \begin{bmatrix} r.3^r & 2^r \\ 0 & (r-1)3^r \end{bmatrix}$ be given matrices.

If $\sum_{r=1}^{50} \text{tr}.((AB)^r C_r) = 3 + a.3^b$ where tr.(A) denotes trace of matrix A, then find the value of $\frac{1}{20}(a + b)$.

[Where a and b are relatively prime]

39. If $\Delta(x) = \begin{vmatrix} x-2 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+2)^3 \end{vmatrix}$, then the absolute value of coefficient of x in $\Delta(x)$ is

40. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, be a 2×2 matrix where $a, b, c, d \in \{0, 1\}$. The number of such matrices which have inverse is

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41. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 9 & 1 & 0 \\ 27 & 9 & 1 \end{bmatrix}$ and $Q = [q_{ij}]_{3\times3}$ be such that $P^5 - Q = I$, then $\dfrac{q_{21} + q_{31}}{q_{32}}$ is equal to

42. Suppose a matrix A satisfies $A^2 - 5A + 7I = O$. If $A^8 = aA + bI$, then a/253 is

43. Let the inequality $\sin^2 x + a \cos x + a^2 \geq 1 + \cos x$ is satisfied $\forall x \in R$, for $a \in (-\infty, k_1] \cup [k_2, \infty)$ then $|k_1| + |k_2| =$

ANSWER KEY

1.	(B)	**2.**	(A)	**3.**	(C)	**4.**	(C)	**5.**	(C)	**6.**	(D)	**7.**	(ABD)
8.	(BD)	**9.**	(BCD)	**10.**	(ABC)	**11.**	(AD)	**12.**	(ACD)	**13.**	(ABC)	**14.**	(ABC)
15.	(ABCD)			**16.**	(ABCD)			**17.**	(AC)	**18.**	(BCD)	**19.**	(BC)
20.	(AB)	**21.**	(AC)	**22.**	(ABC)	**23.**	(AD)	**24.**	(ABCD)			**25.**	(A,B,C)
26.	(A,B,C)			**27.**	(A) $\rightarrow$ (q), (B) $\rightarrow$ (r), (C) $\rightarrow$ (p), (D) $\rightarrow$ (s)								
28.	(A) $\rightarrow$ (q), (B) $\rightarrow$ (r), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)							**29.**	(256)	**30.**	(2)	**31.**	(6)
32.	(16)	**33.**	(7)	**34.**	(1)	**35.**	(65)	**36.**	(31)	**37.**	(2)	**38.**	**(5)**
39.	(2)	**40.**	(6)	**41.**	(22)	**42.**	(5)	**43.**	(3)				

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